

Algebra II with Trigonometry

Chapter 6.5

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Here is a comprehensive breakdown of the provided document, structured to help you understand and master the material.

1. Summary of the Document

This document covers Section 6.5: Inverse Trigonometric Functions and Their Graphs. It introduces the inverse functions for sine, cosine, and tangent by defining their restricted domains and resulting ranges. It details the mathematical notation (e.g., $\sin^{-1} x$ or $\arcsin x$), graphs the functions, and provides the specific cancellation properties for evaluating composite trig functions. The document also briefly outlines the domain restrictions required to define the inverse secant, cosecant, and cotangent functions. Finally, it provides a worksheet of examples ranging from evaluating exact values to rewriting trigonometric compositions as algebraic expressions.

2. Core Definitions, Theorems, and Formulas

Core Definitions

- **Inverse Sine (arcsin):** $y = \sin^{-1} x \iff \sin y = x$.
- Domain: $[-1, 1]$
- Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- **Inverse Cosine (arccos):** $y = \cos^{-1} x \iff \cos y = x$.
- Domain: $[-1, 1]$
- Range: $[0, \pi]$
- **Inverse Tangent (arctan):** $y = \tan^{-1} x \iff \tan y = x$.
- Domain: All real numbers ($\mathbb{R}$)

- Range: $(-\frac{\pi}{2}, \frac{\pi}{2})$
- **Inverse Reciprocal Functions:** To define inverses for secant, cosecant, and cotangent, their original domains are restricted to sets where they are strictly one-to-one. Based on the provided graphs:
- $\sec^{-1} x$ has a range of $[0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$
- $\csc^{-1} x$ has a range of $(0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}]$
- $\cot^{-1} x$ has a range of $(0, \pi)$

Theorems: Cancellation Properties These formulas allow you to "cancel" a trig function with its inverse, provided the input falls within the correct restricted interval.

- **Sine:**
- $\sin(\sin^{-1} x) = x$ for $-1 \leq x \leq 1$
- $\sin^{-1}(\sin x) = x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
- **Cosine:**
- $\cos(\cos^{-1} x) = x$ for $-1 \leq x \leq 1$
- $\cos^{-1}(\cos x) = x$ for $0 \leq x \leq \pi$
- **Tangent:**
- $\tan(\tan^{-1} x) = x$ for $x \in \mathbb{R}$
- $\tan^{-1}(\tan x) = x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

3. Worked Study Guide

To master this material, you need to understand how to apply the definitions and cancellation properties to solve the examples on the final page of the document. Here is a step-by-step guide to the core concepts.

Concept A: Finding Exact Values of Inverse Trig Functions When asked to evaluate an expression like $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$, you are being asked: "*What angle y in the interval $[0, \pi]$ gives a cosine ratio of $-\frac{\sqrt{3}}{2}$?*" 1. Identify the reference angle: We know $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$. 2. Check the sign and range: The ratio is negative. Cosine is negative in Quadrants II and III. However, the range of $\cos^{-1} x$ is strictly $[0, \pi]$ (Quadrants I and II). 3. Find the Quadrant II angle: $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. 4. Result: $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) = \frac{5\pi}{6}$.

Concept B: Using Cancellation Properties (Beware the Traps!)

Sometimes you are given a composition like $\cos^{-1}(\cos(\frac{7\pi}{6}))$.

- *The Trap:* You might be tempted to just cancel the cosines and say the answer is $\frac{7\pi}{6}$.
- *The Rule:* Look at the cancellation properties! $\cos^{-1}(\cos x) = x$ *only* if $0 \leq x \leq \pi$. The angle $\frac{7\pi}{6}$ is larger than π , so you cannot just cancel.
- *The Fix:* Evaluate from the inside out.

1. First, find $\cos(\frac{7\pi}{6}) = -\frac{\sqrt{3}}{2}$. 2. Next, find $\cos^{-1}(-\frac{\sqrt{3}}{2})$, which we found in Concept A to be $\frac{5\pi}{6}$. 3. Result: $\cos^{-1}(\cos(\frac{7\pi}{6})) = \frac{5\pi}{6}$.

Concept C: Evaluating when the Value is Undefined Consider $\sin(\sin^{-1} 5)$.

- Look at the cancellation property: $\sin(\sin^{-1} x) = x$ *only* if $-1 \leq x \leq 1$.
- Because 5 is outside the domain of the inverse sine function (you cannot have a sine ratio greater than 1), $\sin^{-1}(5)$ is **undefined**. The whole expression is undefined.

Concept D: Rewriting as an Algebraic Expression How do you rewrite an expression like $\cos(\sin^{-1} x)$ without trig functions? Use right triangle trigonometry!

1. Let $\theta = \sin^{-1} x$. By definition, this means $\sin \theta = x$. 2.

Write x as a fraction: $\sin \theta = \frac{x}{1} = \frac{\text{Opposite}}{\text{Hypotenuse}}$. 3. Draw a right triangle with

angle θ . Label the opposite side x and the hypotenuse 1. 4. Use the

Pythagorean theorem to find the adjacent side:

$\text{adj}^2 + x^2 = 1^2 \implies \text{adj} = \sqrt{1 - x^2}$. 5. Now evaluate the outside function,

$\cos \theta$. Cosine is $\frac{\text{Adjacent}}{\text{Hypotenuse}}$. 6. Result: $\cos(\theta) = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$.

4. Practice Problems

These problems are pulled directly from the unworked examples provided on page 3 of the document.

Problem 1: Find the exact value: $\tan^{-1} 1$ **Problem 2:** Find the exact value: $\sin^{-1}(-1)$ **Problem 3:** Find the exact value, if defined: $\sin(\sin^{-1} 5)$

Problem 4: Rewrite the expression as an algebraic expression in x : $\cos(\sin^{-1} x)$ **Problem 5:** Find functions f and g such that $F = f \circ g$, given $F(x) = (\tan^{-1} x)^2$.

Answers to Practice Problems:

1. $\frac{\pi}{4}$ (Because $\tan(\frac{\pi}{4}) = 1$ and $\frac{\pi}{4}$ is in the range $(-\frac{\pi}{2}, \frac{\pi}{2})$).
2. $-\frac{\pi}{2}$ (Because $\sin(-\frac{\pi}{2}) = -1$ and $-\frac{\pi}{2}$ is in the range $[-\frac{\pi}{2}, \frac{\pi}{2})$).
3. **Undefined** (Because 5 is not within the domain of $[-1, 1]$ for the inverse sine function).
4. $\sqrt{1 - x^2}$ (See "Concept D" in the study guide above for the step-by-step right triangle method).
5. $g(x) = \tan^{-1} x$ and $f(x) = x^2$ (Because $f(g(x)) = f(\tan^{-1} x) = (\tan^{-1} x)^2 = F(x)$).

5. Assumptions and Ambiguities Resolved

- **OCR/Formatting Glitches:** The raw text (OCR) output missed several mathematical symbols present in the document images, such as fraction bars (e.g., $-\frac{\sqrt{3}}{2}$ became $-3/2$), missing trig function prefixes, and lost exponents. I relied exclusively on the visual screenshots to ensure the correct values and equations were used in this guide.
- **Missing Answer Key:** Page 3 of the document is a blank worksheet. I assumed the role of the tutor was to supply the pedagogical steps required to solve those specific problems, and utilized a selection of those blank examples to create the practice section with a provided answer key.
- **Secant/Cosecant Domains:** Different textbooks handle the range of inverse secant and inverse cosecant differently. I resolved this by carefully reading the specific graphs provided on page 2 of the document. The author strictly defines the range of $\sec^{-1} x$ as $[0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$ and $\csc^{-1} x$ as $(0, \frac{\pi}{2}] \cup (\pi, \frac{3\pi}{2}]$. I assumed this specific mathematical convention must be adhered to for this course.